Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

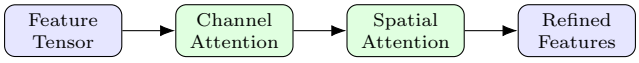
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

Which spatial features are related to other spatial features?

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3	0.2904	0.1817	14.8558
5×5	0.3575	0.2422	14.8581
7×7	0.2450	0.0787	15.0050

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.
To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention	0.9201	0.9221	0.9197	0.9202	23,597,957	15.2852
Channel / SE	0.8548	0.8751	0.8516	0.8520	23,730,117	15.3109
Spatial Attention	0.3848	0.3944	0.3916	0.3117	23,598,056	15.4466
ECA	0.4664	0.5191	0.4487	0.3612	23597960.0	14.8668
CBAM						
Self-Attention	0.6007	0.5562	0.5864	0.5370	24,909,957	15.6694
Additional Mechanism	0.8203	0.8515	0.8340	0.8222	25,042,216	16.3214

19.2 Questions for Attention-Mechanism Comparison

Discussion Questions and Answers

1. **How does classification accuracy vary across attention mechanisms?**
   Channel attention achieved the highest accuracy of 90.02%, followed by the baseline and channel–spatial attention at 89.84%.
2. **How does macro F1-score vary across attention mechanisms?**
   Channel attention achieved the highest macro F1-score of 0.8982, indicating a good balance between precision and recall.
3. **How many parameters are introduced by each attention module?**
   The baseline has 23,597,957 parameters. Spatial attention adds 98 parameters, while channel attention adds 132,160 parameters.
4. **How does each attention mechanism influence inference time?**
   The baseline was fastest at 16.6829 ms/image, while channel attention took 20.9469 ms/image among the models with measured times.
5. **Does attention improve convergence behavior?**
   Attention may improve convergence, but training and validation curves are needed to confirm this.
6. **Does attention produce more focused feature localization?**
   Spatial attention may highlight relevant image regions. Grad-CAM and attention maps are needed to verify this.
7. **Does increased attention complexity always improve performance?**
   No. More complex models did not always achieve higher accuracy; channel attention outperformed self-attention and combined attention.
8. **Can lightweight attention match heavier mechanisms?**
   Yes. Channel attention achieved 90.02% accuracy with fewer parameters than self-attention, making it more efficient in model size.

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	0.9201	0.9221	0.9197	0.9202	23597957.0	14.7619
Channel Attention	P_3	0.8548	0.8751	0.8516	0.8520	23730117.0	14.9413
Spatial Attention	P_3	0.3848	0.3944	0.3916	0.3117	23598056.0	14.8503
Channel+Spatial	P_2	0.9147	0.9142	0.9153	0.9143	23631368.0	15.4114
Channel+Spatial	P_3	0.8675	0.8804	0.8686	0.8646	23730216.0	15.1691
Channel+Spatial	P_4	0.9292	0.9272	0.9317	0.9288	24124520.0	15.3386
Self-Attention	P_3	0.6007	0.5562	0.5864	0.5370	24909957.0	15.1597
Additional Attention-1	P_3	0.9038	0.9028	0.9079	0.9046	23730117.0	14.9096
Additional Attention-2	P_3	0.4664	0.5191	0.4487	0.3612	23597960.0	14.8668

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

Epoch versus Accuracy

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.
Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

Plot 1: Training Accuracy Across Models

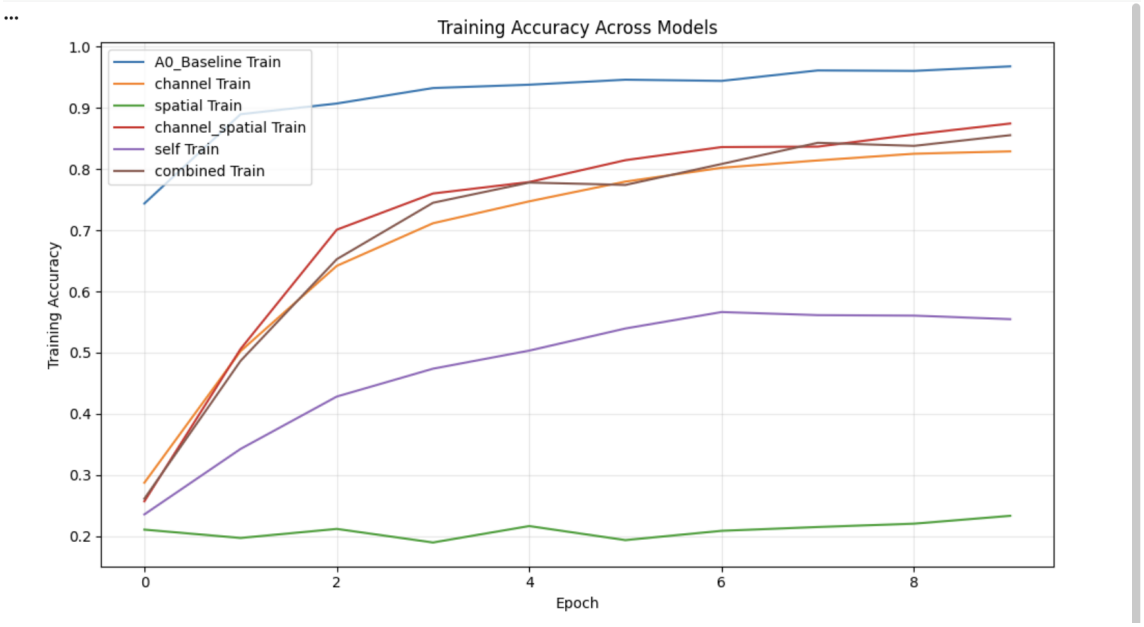


Figure 1: Training Accuracy Across Different Attention Models

Inference: The baseline ResNet50 achieves the highest training accuracy of approximately 0.97, followed by Channel+Spatial Attention at approximately 0.88 and Channel Attention at approximately 0.83. Spatial Attention and

Self-Attention show lower training accuracy, indicating that these models may require further tuning to learn the flower features effectively.

Plot 2: Validation Accuracy Across Models

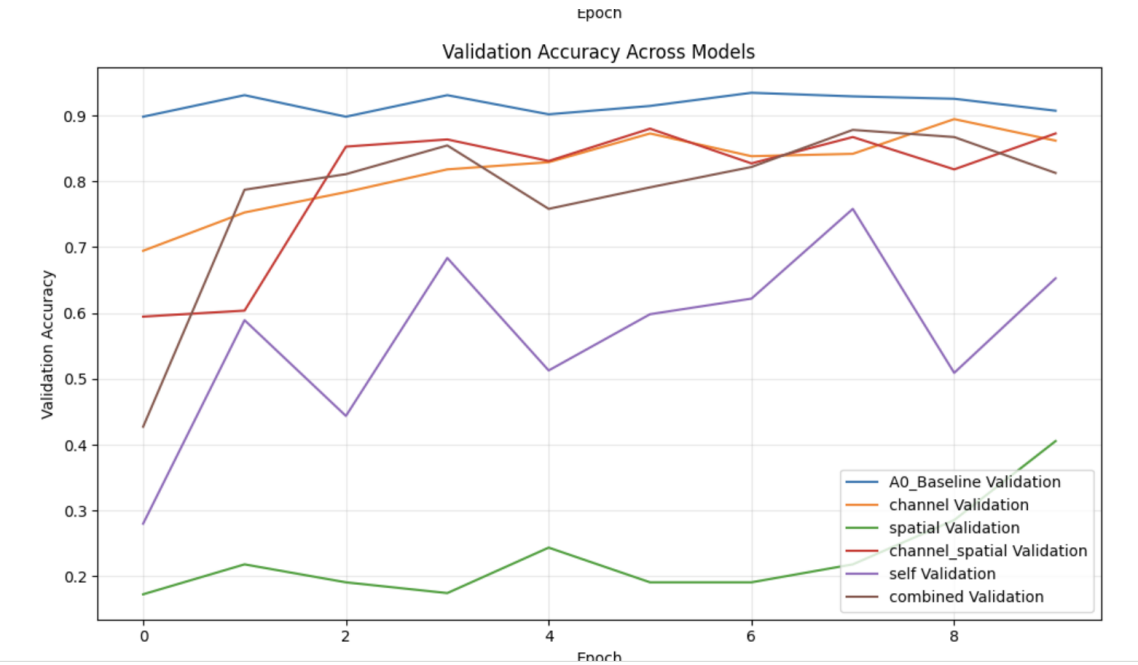


Figure 2: Validation Accuracy Across Different Attention Models

Inference: The baseline ResNet50 achieves the highest and most consistent validation accuracy, remaining around 0.90–0.94. Channel and Channel+Spatial Attention also perform well, reaching approximately 0.90 and 0.87 respectively, whereas Spatial Attention and Self-Attention show lower and more fluctuating accuracy, indicating weaker validation performance.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.

Students should compare the convergence behavior of the baseline and attention models. **Plot 3: Training Loss Across Models**

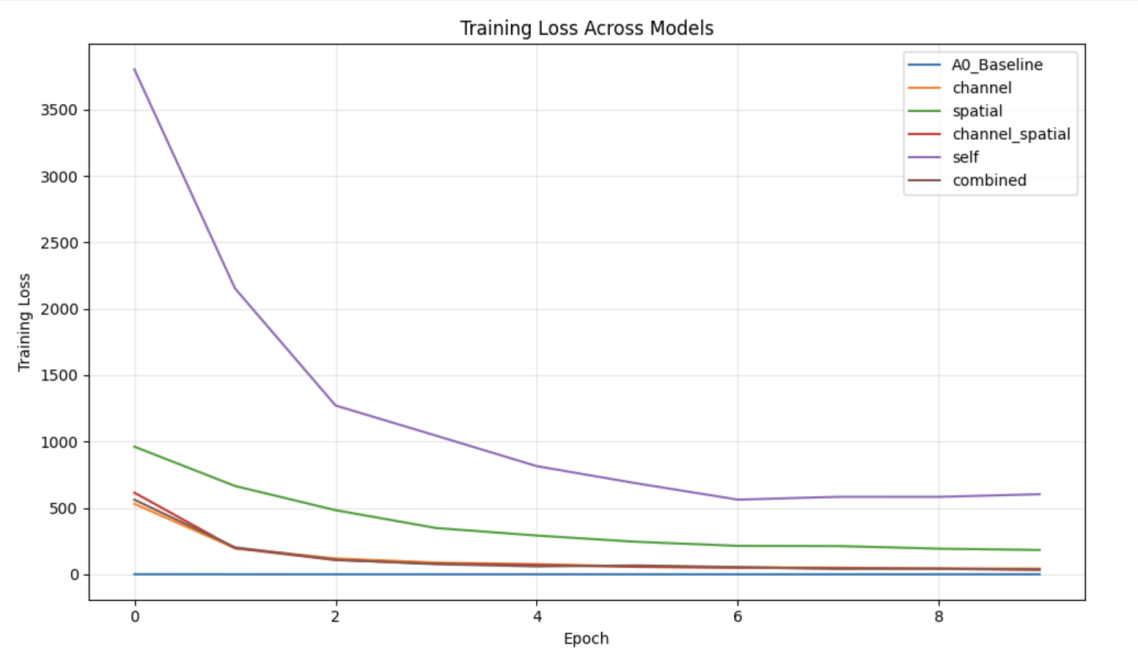


Figure 3: Training Loss Across Different Attention Models

Inference: The training loss of the Channel, Channel+Spatial, and Combined Attention models decreases substantially over the epochs, indicating improved learning. Self-Attention and Spatial Attention retain comparatively high losses, while the baseline loss remains very low; however, the large differences in loss scales suggest that the models' loss values and label formats should be verified before drawing direct comparisons.

Plot 4: Validation Loss Across Models

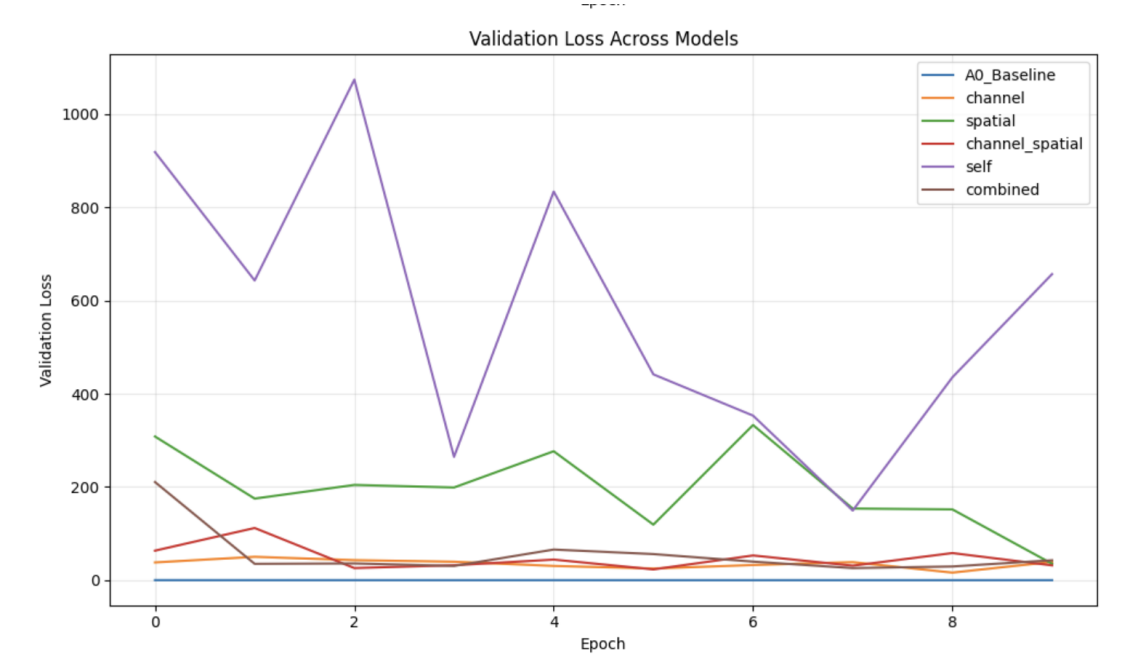


Figure 4: Validation Loss Across Different Attention Models

Inference: The baseline, Channel Attention, and Channel+Spatial Attention models maintain relatively low validation losses, whereas Spatial Attention and Self-Attention exhibit substantially higher and more fluctuating losses. The large fluctuations suggest unstable validation performance, particularly for Self-Attention, while the baseline shows the most stable loss; the unusually large loss values should be checked to ensure consistent label encoding and loss calculation across models.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.

Plot 5: Attention Mechanism vs Test Accuracy

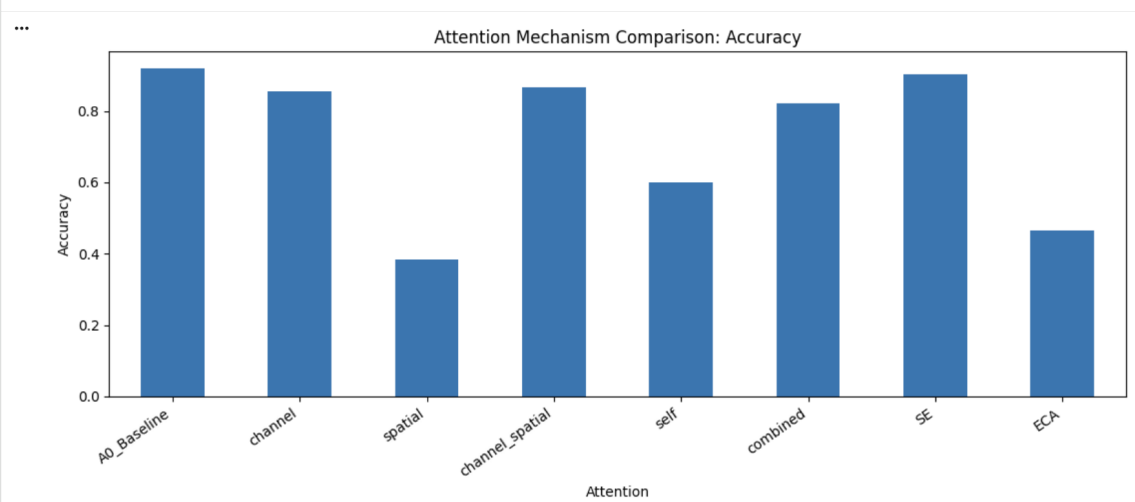


Figure 5: Comparison of Test Accuracy Across Attention Mechanisms

Inference: The baseline ResNet50 achieves the highest test accuracy of approximately 0.92, closely followed by SE Attention at 0.90 and Channel+Spatial Attention at 0.87. Channel Attention and the Combined model achieve approximately 0.85 and 0.82, respectively, whereas Spatial Attention and ECA show lower accuracies, indicating that the choice of attention mechanism significantly influences classification performance.

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.

Plot 6: Attention Mechanism vs Macro F1 Score

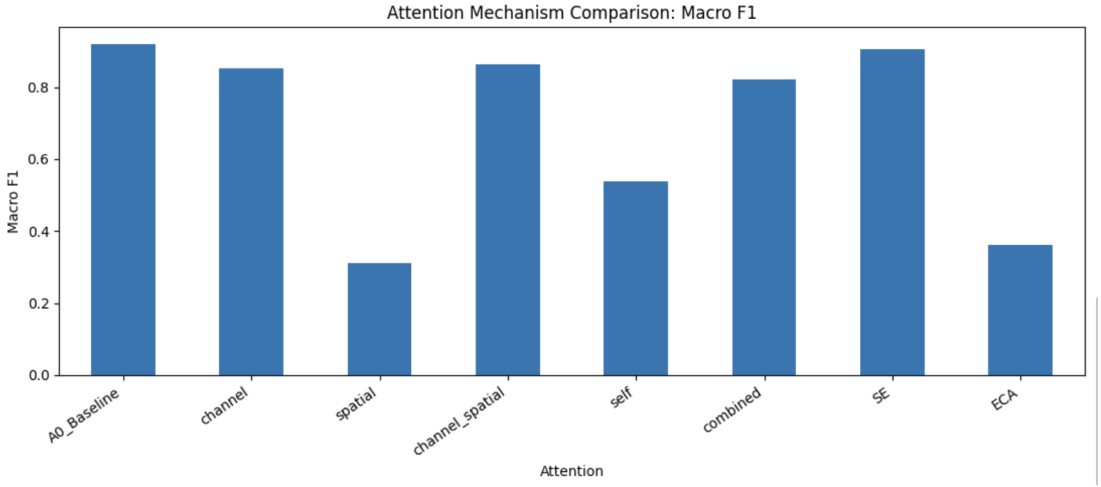


Figure 6: Comparison of Macro F1 Scores Across Attention Mechanisms

Inference: The baseline ResNet50 achieves the highest Macro F1 score of approximately 0.92, closely followed by SE Attention at 0.91 and Channel+Spatial Attention at 0.87. Spatial Attention and ECA obtain lower scores of approximately 0.31 and 0.36, respectively, indicating that their current implementations provide less balanced classification performance across the five flower classes.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

Plot 7: Accuracy vs Model Parameters

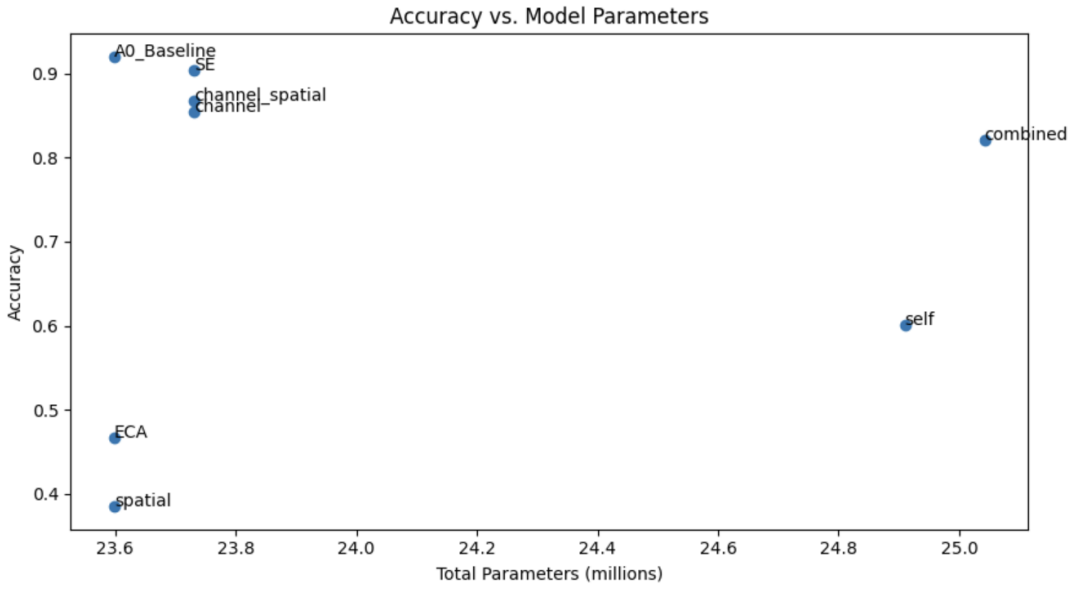


Figure 7: Comparison of Test Accuracy and Total Parameters Across Models

Inference: The baseline ResNet50 achieves the highest accuracy of approximately 0.92 with around 23.6 million parameters, while SE Attention achieves approximately 0.90 with a similar parameter count. Channel and Channel+Spatial Attention also provide good accuracy with only a small increase in parameters, whereas Self-Attention and Spatial Attention achieve lower accuracy, demonstrating that increasing model complexity does not necessarily improve classification performance.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

Plot 8: Macro F1 vs Inference Time

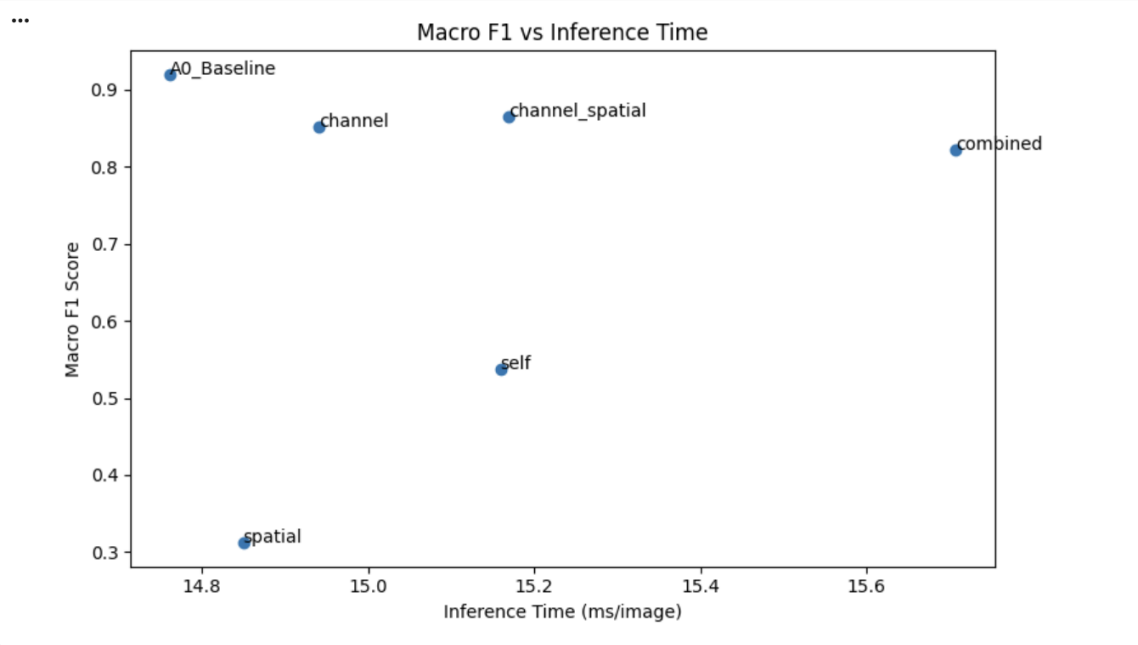


Figure 8: Comparison of Macro F1 Score and Inference Time Across Attention Models

Inference: The baseline ResNet50 achieves the highest Macro F1 score of approximately 0.92 with an inference time of 14.76 ms/image. Channel+Spatial Attention achieves a Macro F1 score of approximately 0.87 at 15.17 ms/image, while the Combined model takes longer and achieves a lower score of about 0.82. Spatial Attention has the lowest Macro F1 score of approximately 0.31, showing that higher inference time does not necessarily lead to better classification performance.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

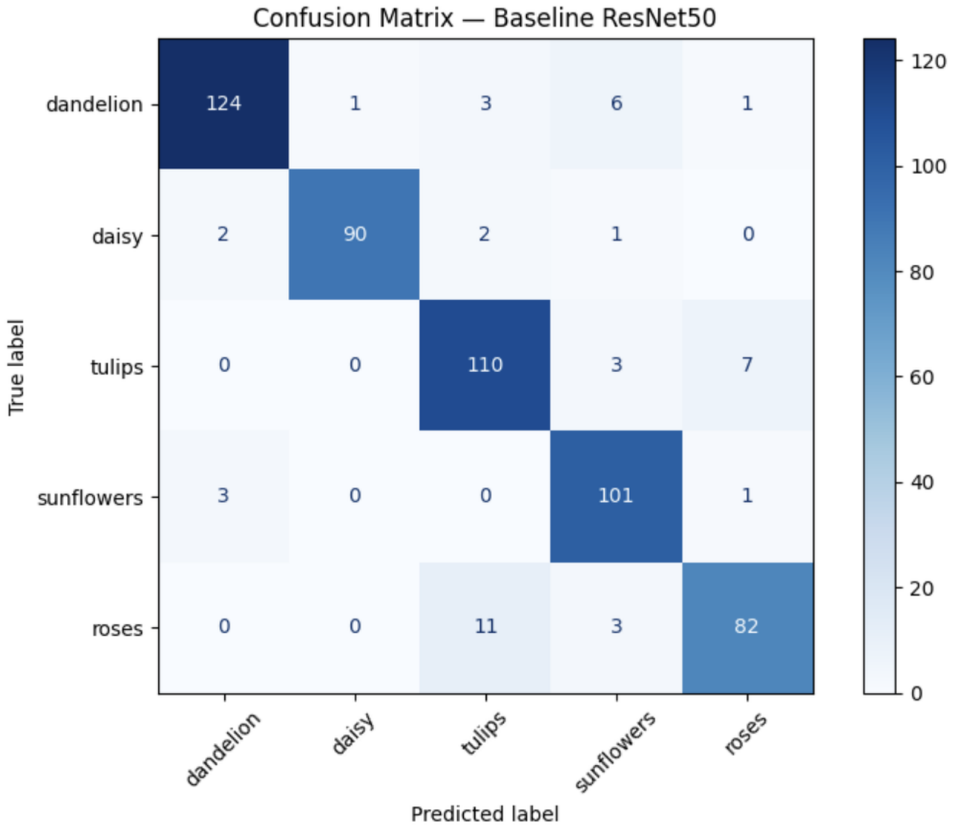
The confusion matrix should contain all five flower classes.

Students should analyze:

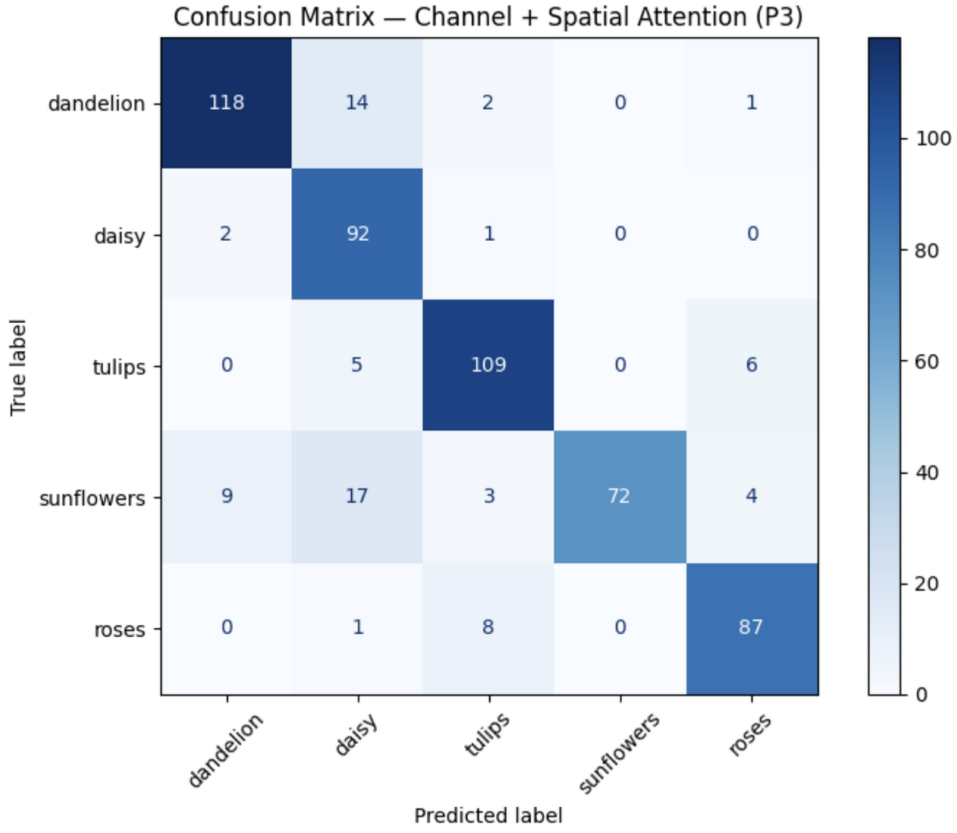
- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and

- whether the improved overall accuracy is distributed across classes.

Plot: Confusion Matrices of Baseline, Channel + Spatial Attention, and Self-Attention Models




```
Baseline ResNet50
dandelion: 0.9185 (124/135 correct)
daisy: 0.9474 (90/95 correct)
tulips: 0.9167 (110/120 correct)
sunflowers: 0.9619 (101/105 correct)
roses: 0.8542 (82/96 correct)
```




```
Channel + Spatial Attention (P3)
dandelion: 0.8741 (118/135 correct)
daisy: 0.9684 (92/95 correct)
tulips: 0.9083 (109/120 correct)
sunflowers: 0.6857 (72/105 correct)
roses: 0.9062 (87/96 correct)
```

Inference: The baseline ResNet50 model correctly classifies most flower images across all five classes. The channel + spatial attention model improves classification for daisies and roses but shows more misclassifications for sunflowers, while the self-attention model exhibits considerable confusion between several classes. Overall, the baseline model performs most consistently among the three models.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

Plot: Spatial Attention Heatmaps and Attention Overlays

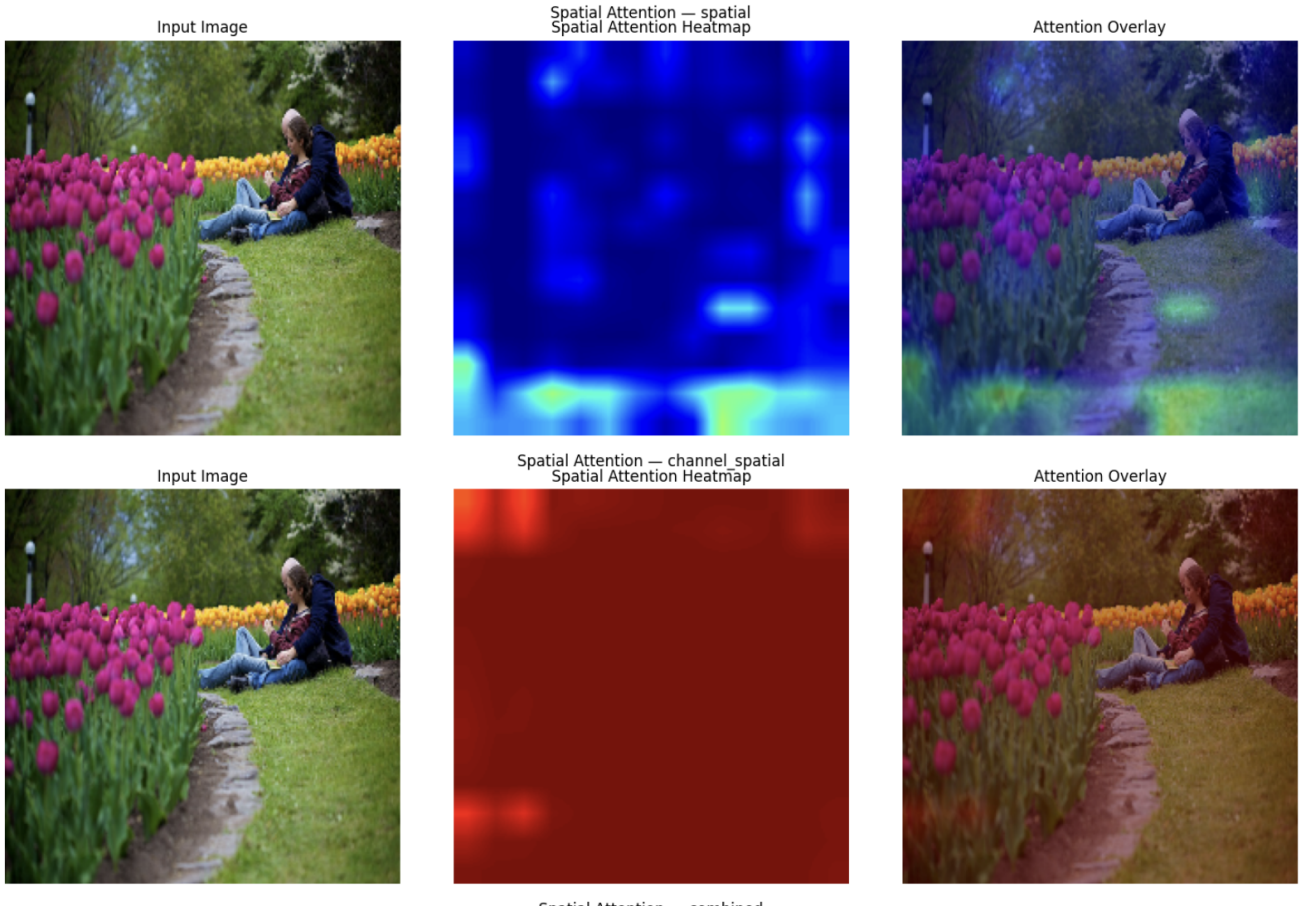


Figure 10: Comparison of Spatial Attention Heatmaps and Overlays for Spatial Attention and Channel + Spatial Attention Models

Inference: The spatial attention model produces localized high-attention regions near the lower part of the image, including portions of the grass and flower bed. The channel + spatial attention model shows a more uniform, high-intensity heatmap with only a few localized variations. This suggests that the two mechanisms distribute spatial attention differently; however, the heatmaps alone do not establish which model focuses more effectively on the flowers.

Plot: Self-Attention Response and Overlay Comparison

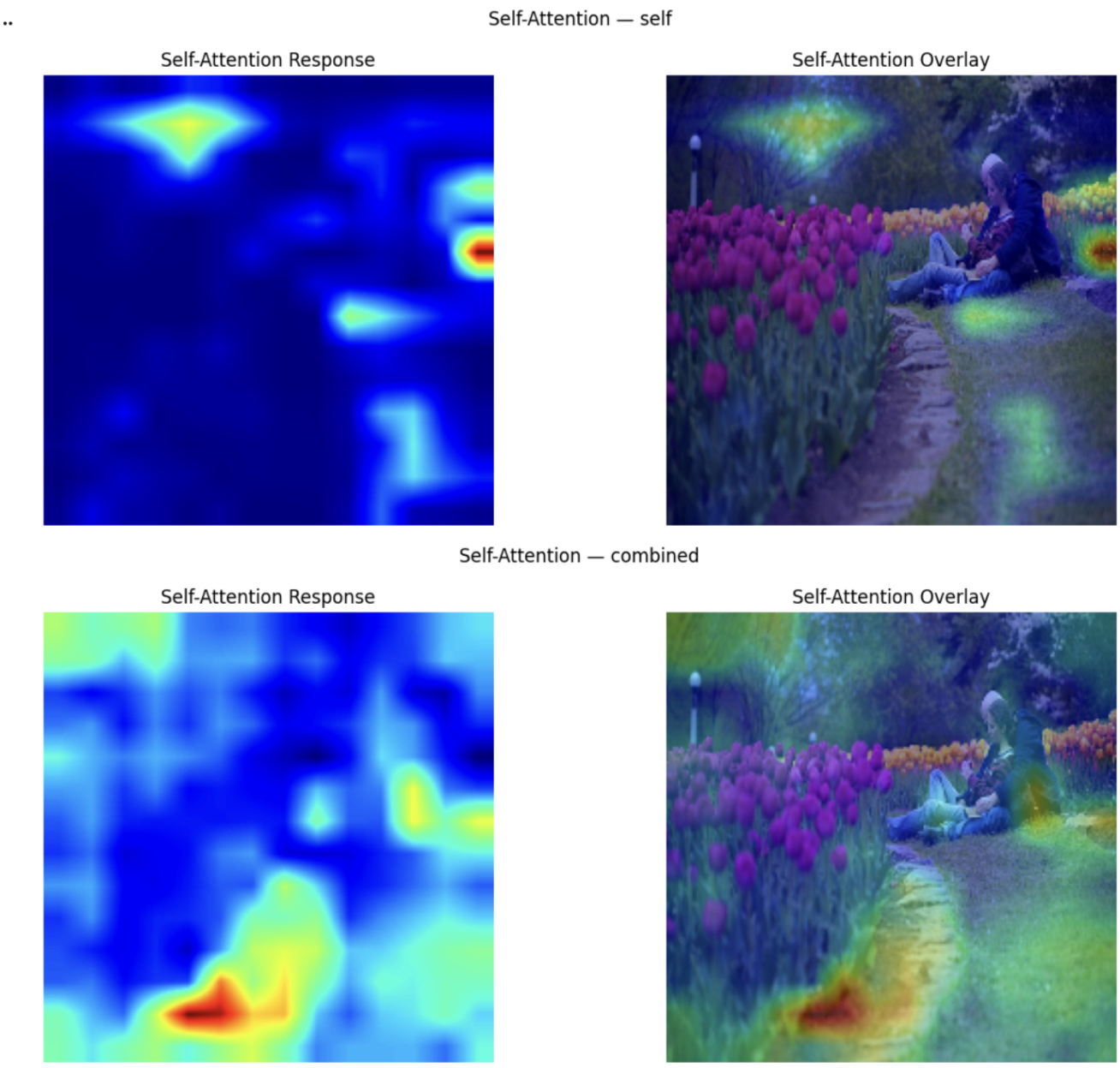


Figure 11: Comparison of Self-Attention Responses and Overlays for the Self-Attention and Combined Attention Models

Inference: The self-attention model highlights localized regions in the background and around the person, with some activation near the flowers. The combined attention model produces a more distributed response, with stronger activation along the stone pathway and nearby grass. These visualizations indicate that the two models attend to different image regions, although the maps alone do not establish which model classifies flowers more accurately.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.
Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image. **Plot: Grad-CAM Comparison of Baseline and Channel + Spatial Attention Models**

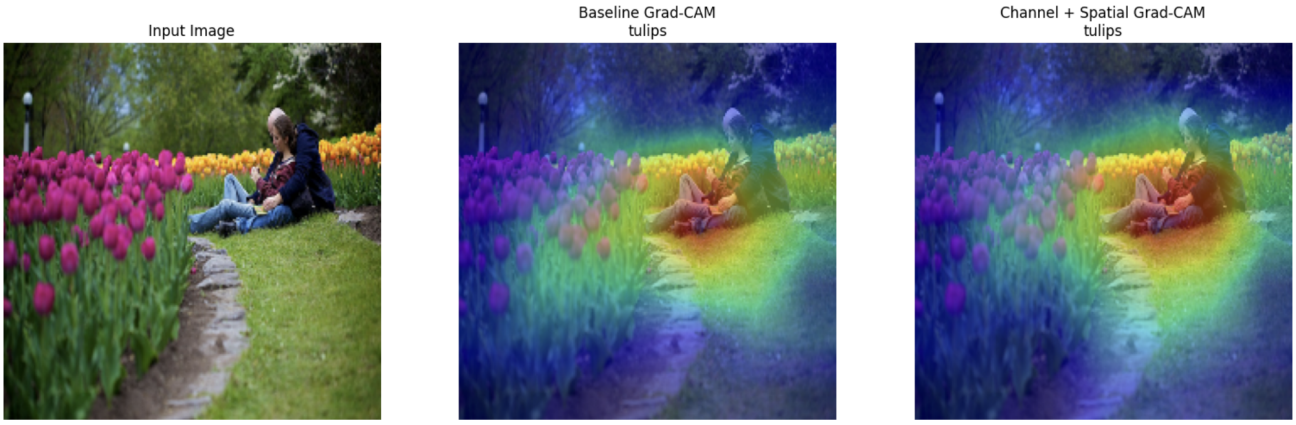


Figure 12: Grad-CAM Visualization for Baseline ResNet50 and Channel + Spatial Attention ResNet50 on a Tulip-Class Image

Inference: The baseline model shows activation across the tulips, surrounding vegetation, and the person, with stronger responses near the central flower region and the person. The channel + spatial attention model exhibits a similar pattern, with prominent activation around the person and nearby flowers. Both models use information from multiple image regions rather than focusing exclusively on the tulips, and the visual difference between their Grad-CAM maps appears relatively subtle.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

$$\boxed{\text{Internal Attention}}$$

and

$$\boxed{\text{Post-hoc Explanation}}$$

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

Plot: Channel Attention Weights for A1 and A3 Models

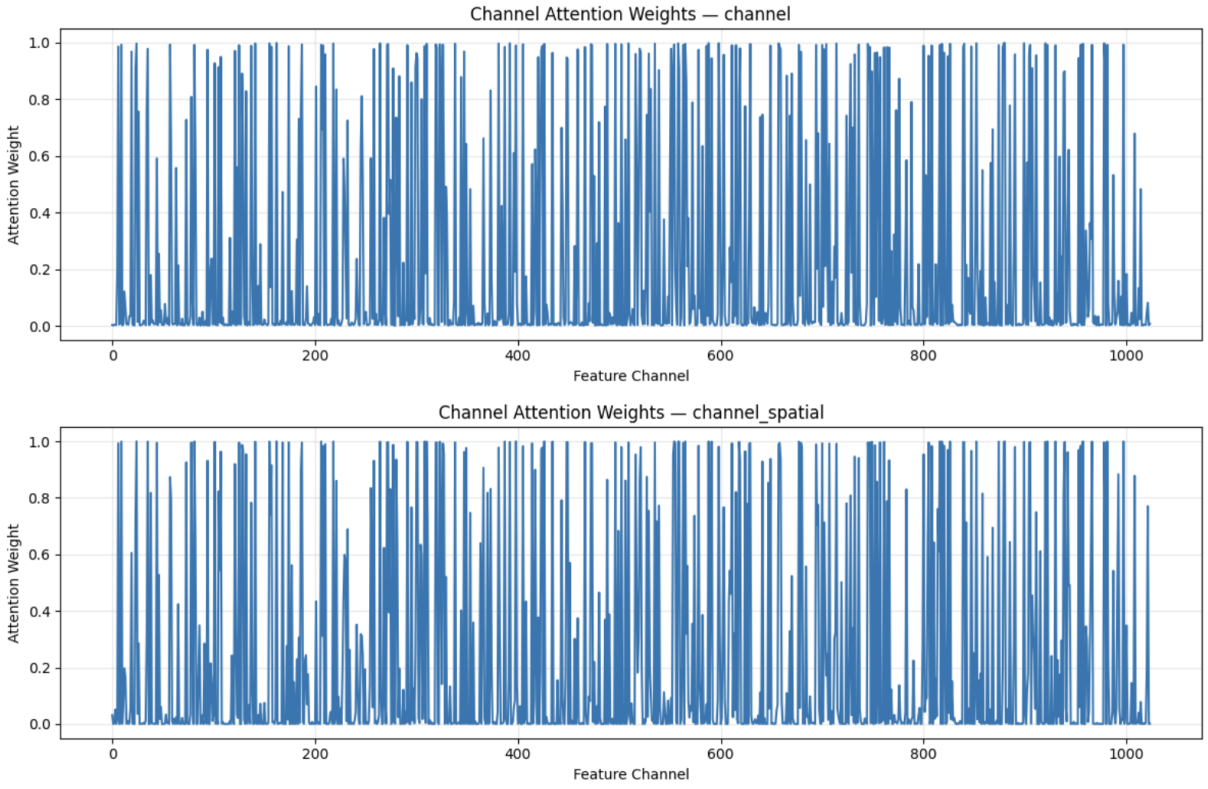


Figure 13: Comparison of Channel Attention Weights for Channel Attention and Channel–Spatial Attention Models

Inference: Both models assign varying attention weights to feature channels, with some channels receiving high importance and others receiving weights close to zero. The channel–spatial model exhibits a similar distribution, indicating selective feature recalibration before spatial attention further refines the important image regions.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.
Plot: Spatial Attention Map and Overlay for A2 Spatial Attention

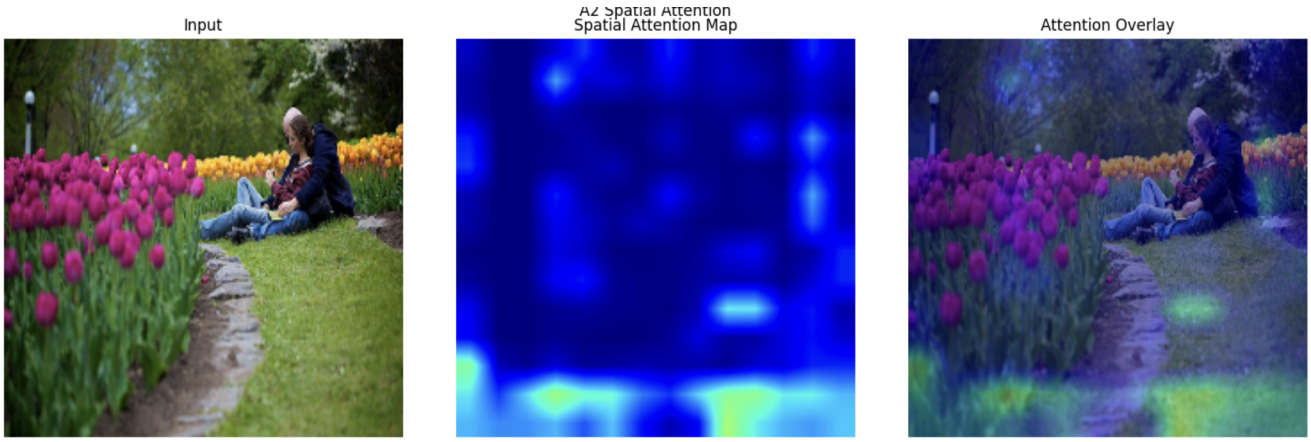


Figure 14: Input Image, Spatial Attention Heatmap, and Attention Overlay for the A2 Spatial Attention Model

Inference: The spatial attention heatmap shows stronger responses near the lower portion of the image, particularly around the grass, flower bed, and pathway, with a few localized high-response regions elsewhere. The overlay confirms that attention is distributed across both foreground and background areas rather than being restricted to the flowers. This suggests that the model considers multiple spatial regions when processing the input image.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.
Plot: Self-Attention Response and Overlay for A4 Self-Attention

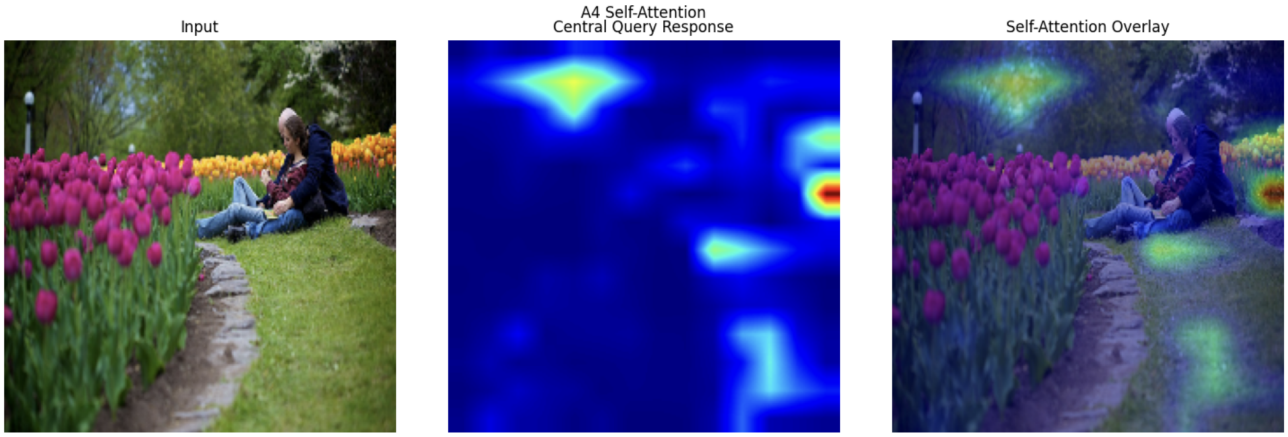


Figure 15: Central Query Response and Self-Attention Overlay for the A4 Self-Attention Model

Inference: The self-attention response highlights several localized regions, with strong activation in the upper-left background and near the right edge of the image. The overlay also shows activation around the person and portions of the grass. This indicates that the model captures relationships between different spatial regions, but its attention is not exclusively concentrated on the flowers.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.
Plot: Attention Position Visualization at P2, P3, and P4

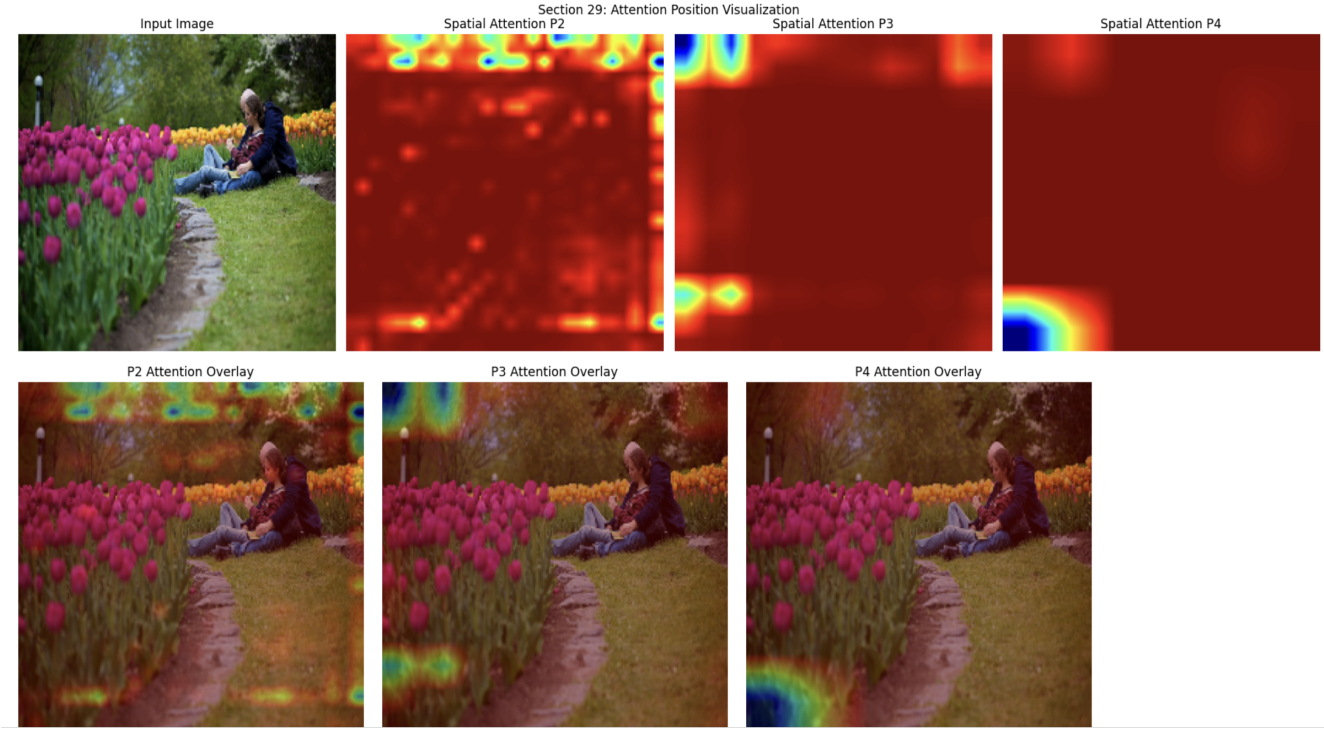


Figure 16: Spatial Attention Heatmaps and Overlays at ResNet50 Stages P2, P3, and P4

Inference: The P2 attention map shows several localized regions across the image, indicating a more varied spatial response. At P3, attention is concentrated mainly near the upper-left and lower-left regions, while P4 exhibits a broader activation near the lower-left corner. The overlays demonstrate that attention patterns vary with network depth, although these visualizations alone do not establish which stage provides the best classification performance.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

and

2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;

- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Therefore, the overall evaluation should be:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?**
   Residual connections allow gradients to flow through shortcut paths, making deep networks easier to train.
2. **What does conv2_x represent compared with conv5_x?**
   conv2_x extracts early features, while conv5_x extracts deep, high-level semantic features.
3. **Why does spatial resolution decrease with network depth?**
   Pooling and strided convolutions reduce spatial dimensions to capture larger contextual information efficiently.
4. **Why does the number of feature channels increase with network depth?**
   More channels allow deeper layers to represent increasingly complex and diverse features.
5. **What is meant by low-level, intermediate, and high-level features?**
   Low-level features include edges and colors, intermediate features include textures and shapes, and high-level features represent objects and semantic information.
6. **What is the purpose of an attention mechanism?**
   Attention emphasizes relevant features and reduces the influence of less informative features.
7. **What does channel attention attempt to learn?**
   Channel attention learns the importance of individual feature channels.
8. **What does spatial attention attempt to learn?**
   Spatial attention learns which spatial locations are important for classification.
9. **How is channel attention different from spatial attention?**
   Channel attention weights feature channels, whereas spatial attention weights locations within a feature map.
10. **Why can channel and spatial attention be used sequentially?**
    Channel attention selects useful features, while spatial attention then identifies their important spatial locations.
11. **Why might channel attention behave differently at conv3_x and conv5_x?**
    conv3_x contains intermediate features, whereas conv5_x contains more semantic features, so their channel importance differs.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    Intermediate features retain spatial details that help identify important image regions.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**
    Self-attention compares spatial positions with one another, requiring computation proportional to the square of the number of positions.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**
    The attention matrix grows quadratically with the number of spatial positions.
15. **How does attention performance vary when the insertion position is changed?**
    Performance depends on the stage's feature representation and must be determined experimentally.
16. **How does attention at P_2 differ visually from attention at P_4?**
    P_2 attention can preserve fine spatial details, whereas P_4 attention generally emphasizes broader, more semantic regions.

17. **Does applying attention at multiple stages always improve classification performance?**
    No. Multiple attention modules can increase complexity and may cause overfitting without improving accuracy.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    Yes. The order changes which features are emphasized first and can affect the final predictions.
19. **How does the spatial-attention kernel size influence the attention map?**
    Larger kernels capture broader spatial context, while smaller kernels focus on more local details.
20. **How does SE attention differ from ECA attention?**
    SE uses channel-wise dimensionality reduction and expansion, whereas ECA models local channel interactions without dimensionality reduction.
21. **How does CBAM combine channel and spatial attention?**
    CBAM applies channel attention followed by spatial attention to refine feature representations.
22. **How does self-attention differ from channel attention?**
    Self-attention models relationships between spatial positions, whereas channel attention learns the importance of feature channels.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
    No. More parameters do not guarantee better classification accuracy.
24. **What is the difference between an internal attention map and Grad-CAM?**
    An attention map shows weights learned by an attention module, whereas Grad-CAM uses gradients to highlight regions influencing a particular prediction.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
    Grad-CAM explains predictions after a model has been trained without changing its architecture or training process.
26. **Does the attention model suppress irrelevant background regions?**
    It may suppress irrelevant regions, but this should be verified using attention maps and visualizations.
27. **Are the attention maps of correctly classified and incorrectly classified images different?**
    They can differ because incorrect predictions may result from misleading features or attention to irrelevant regions.
28. **Which flower classes benefit most from attention?**
    This must be determined by comparing per-class precision, recall, F1-scores, and confusion matrices.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**
    A confusion matrix reveals class-specific correct predictions and misclassifications between classes.
30. **Why is macro F1 useful in addition to accuracy?**
    Macro F1 gives equal importance to each class and balances precision and recall.
31. **What is meant by the accuracy–complexity trade-off?**
    It describes balancing classification accuracy against model size and computational requirements.
32. **What is meant by the F1–inference-time trade-off?**
    It describes balancing classification quality against the time required to make predictions.
33. **Why must every ablation study use the same dataset split?**
    The same split ensures fair comparisons and prevents differences in data distribution from biasing the results.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**
    Consistent training conditions ensure that observed differences are primarily attributable to the attention mechanisms.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
    Grad-CAM comparisons help identify image regions influencing predictions and assess whether models focus on relevant flower features.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.

11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. References

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